

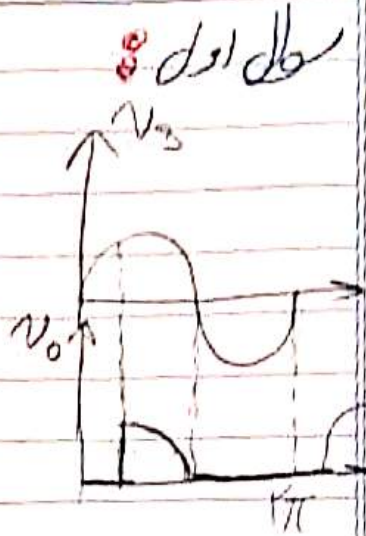
۱۱ "اگر من بفرماید که من ۱۲۵۲ هجری را میگویم"

12.051

Subject:

$PF = P \rightarrow$ طاب
 $S \rightarrow$ لما ظلمني

$$P = \frac{V_{rms}^2}{R}$$
$$S = V_{in, rms} \times I_{in, rms}$$



$$V_{G_{rms}} = \sqrt{\frac{1}{\pi} \int_0^{\pi} [V_m \sin(\omega t)]^2 d\omega t} = \frac{V_m}{\sqrt{\pi}} \sqrt{\int_0^{\pi} \frac{1 - \cos 2\omega t}{2} d\omega t}$$

$$= \frac{V_m}{\sqrt{2\pi}} \sqrt{\frac{\omega t - \frac{1}{\pi} \sin(\pi \omega t)}{r}} - \frac{V_m}{\sqrt{2\pi}} \sqrt{\frac{(\pi - \alpha) - \frac{1}{\pi} (\sin 2\pi - \sin 2\alpha)}{r}}$$

$$\Rightarrow P = \frac{V_o^2 R_{ms}}{R} = \frac{V_m^2}{4R} \times \frac{\pi - \alpha + \frac{1}{\gamma} \sin \alpha}{\gamma R}$$

$$V_{n,RMS} = V_{s,RMS} = \frac{V_m}{\sqrt{2}}$$

$$I_{in, RMS} \rightarrow I_{in} = I_o \Rightarrow I_{in, RMS} = I_{o, RMS} = \frac{V_{o, RMS}}{R} \times \frac{V_{o, RMS}}{R}$$

$$\Rightarrow PF = \frac{P}{S} = \frac{V_{o,RMS} \times \frac{1}{\frac{V_m}{\sqrt{2}} \times \frac{V_{o,RMS}}{R}}}{\frac{V_m}{\sqrt{2}} \times \frac{V_{o,RMS}}{R}} = \frac{V_{o,RMS}}{V_m} \sqrt{2} = \frac{1}{\sqrt{2}} \sqrt{1 - \sin^2 \alpha}$$

$$= \sqrt{\frac{1}{\epsilon} - \frac{\alpha}{ra} + \frac{1}{\epsilon a} \sin(\alpha r)} \quad \checkmark$$

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$$P_{net} = 2 \text{ kw}$$

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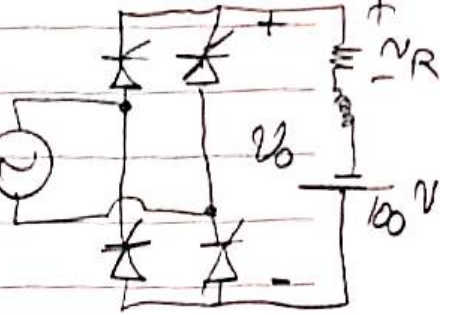
$$P_{solar} = P_R + P_{net} \Rightarrow V_o \bar{I}_o = 0.1 \bar{I}_{o, RMS} \text{ kw} \Rightarrow 100 \bar{I}_o = 0.1 \bar{I}_o$$

$$I_o: DC$$

$$\Rightarrow \bar{I}_o = 10 \text{ A}$$

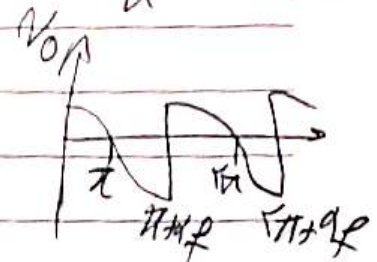
$$\bar{V}_o = V_R - V_{solar} = 0.1 \times 100 - 100 = -10$$

20 V



$$\bar{V}_o = \frac{1}{\pi} \int_{\pi+\alpha_f}^{\pi+\alpha_f} -V_m \sin(\omega t + \alpha_f) d\omega t = \frac{V_m}{\pi} \times \cos(\alpha_f) \Rightarrow \frac{150\sqrt{2}}{\pi} \times \cos(\alpha_f) = -10$$

$$\Rightarrow \alpha_f = 151^\circ$$

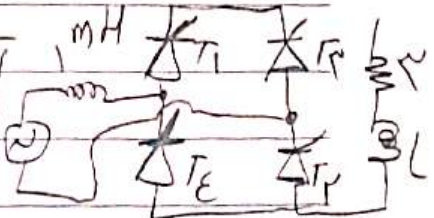


$$P_{solar} = P_R + P_{net} = 2 \text{ kw} + 0.1 \times 100 = 2.1 \text{ kw}$$

ج

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حرف فزوی DC $\Rightarrow$ تبدیل خلی علم $\Rightarrow$ استخراج

$$\Rightarrow P_{out} = P_R = R I_{out}^2 = 1 \text{ kw} \Rightarrow I_{out} = \sqrt{\frac{1 \text{ kw}}{R}} = 2.45 \text{ A}$$



Shahab

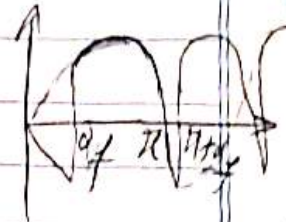
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$$I_{out} = \frac{V_{out}}{R} \rightarrow \text{در حالت DC، ولتاژ مدار} \Rightarrow \frac{V_{out}}{R} = 21.4 \Rightarrow V_{out} = 21.4 \times R$$

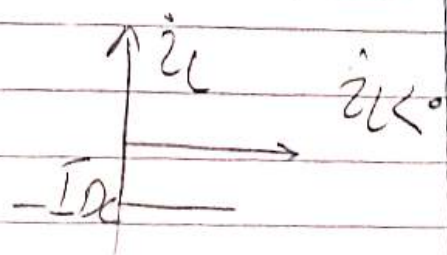
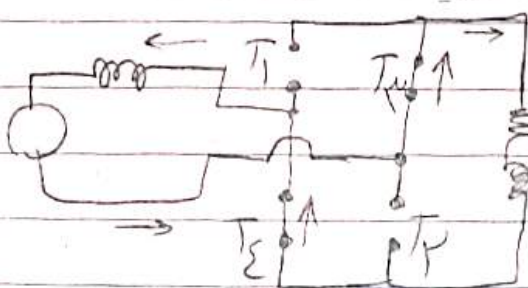
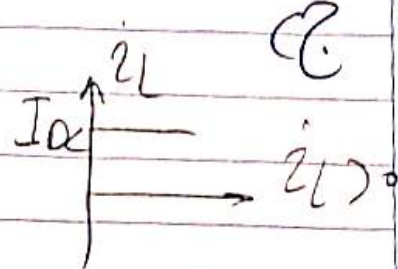
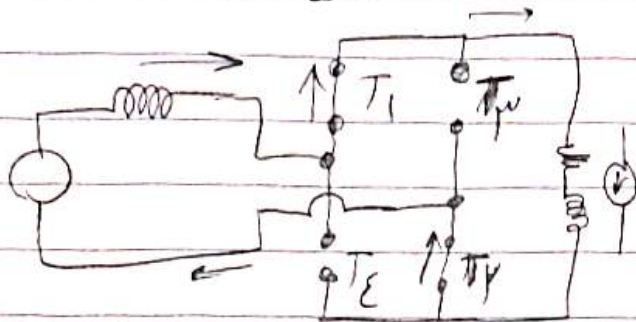
امید من بلف در یک منبع تغذیه است که می تواند کار کند.

$$V_{out} = \frac{1}{\pi} \int_{\alpha f}^{\alpha f + \pi} V_m \sin(\omega t) d\omega t = \frac{V_m \sqrt{2}}{\pi} (\cos \alpha f - \cos(\alpha f + \pi))$$

$$\Rightarrow \frac{V_m \sqrt{2}}{\pi} \times 2 \cos(\alpha f) = 21.4 \Rightarrow \alpha f = 49^\circ$$



نقشه مدار متاد



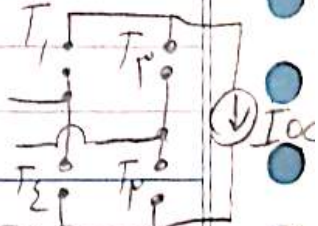
نقشه مدار متاد



① kcl: $I_{T1} + I_{T3} = I_{DC}$

② //: $I_{T1} + I_{T3} = I_{DC}$

③ بهای جاری در هر لحظه از زمان
 (تفاوت ضریب در هر لحظه)



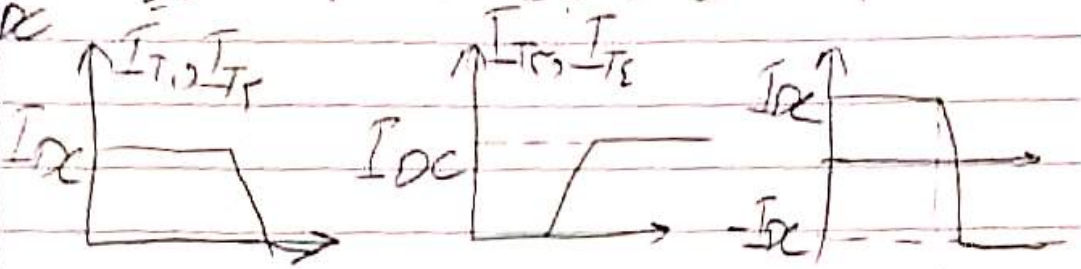
محل

③ ② ①: در باره تغییر شکل سلف، هر دو منبع بار و هر دو منبع بار و هر دو منبع بار



بنا بر این در باره تغییر شکل سلف، هر دو منبع بار و هر دو منبع بار و هر دو منبع بار

در تیری به آبر که در همین باره، هر دو منبع بار و هر دو منبع بار و هر دو منبع بار



$$\Rightarrow v_L = v_{in} = L \frac{di_L}{dt} \Rightarrow \frac{v_{in}}{L} dt = \frac{v_{in} di_L}{L v_{in}} \Rightarrow di_L = \frac{v_{in}}{L} dt$$

$$= di_L \xrightarrow{\text{انتگرال گیری}} \int_{-\pi/2}^{\pi/2} v_{in} \cos(\omega t) dt = \int_{-I_{DC}}^{+I_{DC}} di_L \Rightarrow \int_{-\pi/2}^{\pi/2} \frac{V_m}{L} \sin(\omega t) dt = \int_{-I_{DC}}^{+I_{DC}} di_L$$

$$\Rightarrow -\frac{V_m}{L} \cos(\omega t) \Big|_{-\pi/2}^{\pi/2} = -I_{DC} \Rightarrow -\frac{V_m}{L} [\cos(\pi/2 + \pi) - \cos(\pi/2 - \pi)] = -I_{DC}$$

$$= -I_{DC} \Rightarrow \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi) = I_{DC} \Rightarrow u = -\pi - \pi/2 + \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi)$$

$$\Rightarrow u = -\pi - \pi/2 + \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi) = -\pi - \pi/2 + \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi)$$

$$\Rightarrow u = -\pi - \pi/2 + \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi) = -\pi - \pi/2 + \cos(\pi/2 + \pi) + \cos(\pi/2 - \pi)$$

